

§2 Trigonometric Integrals and Substitution

* Trigonometric Integrals

Exam 1 Evaluate $\int \cos^3 x \, dx$

$$\begin{aligned} \text{(sol) I. } \int \cos^3 x \, dx &= \int \cos^2 x \cos x \, dx \\ &= \cos^2 x \sin x + \int 2\cos x \sin x \sin x \, dx \\ &= \cos^2 x \sin x + 2 \int \cos x \sin^2 x \, dx \\ &= \cos^2 x \sin x + 2 \int \cos x (1 - \cos^2 x) \, dx \\ &= \cos^2 x \sin x + 2 \int \cos x \, dx - 2 \int \cos^3 x \, dx \end{aligned}$$

$$\begin{aligned} \Rightarrow 3 \int \cos^3 x \, dx &= \cos^2 x \sin x + 2 \int \cos x \, dx \\ &= \cos^2 x \sin x + 2 \sin x + 3C \end{aligned}$$

$$\therefore \int \cos^3 x \, dx = \frac{1}{3}(\cos^2 x + 2)\sin x + C$$

II. using $\cos 3x = 4\cos^3 x - 3\cos x$, we get

$$\begin{aligned} \int \cos^3 x \, dx &= \int \left(\frac{1}{4} \cos 3x + \frac{3}{4} \cos x \right) dx \\ &= \frac{1}{12} \sin 3x + \frac{3}{4} \sin x + C \end{aligned}$$

$$\begin{aligned} \text{III. } \int \cos^3 x \, dx &= \int \cos^2 x \cos x \, dx \\ &= \int (1 - \sin^2 x) \cos x \, dx \\ &= \int \cos x \, dx - \int \sin^2 x \cos x \, dx \\ &= \int \cos x \, dx - \int \sin^2 x \, d(\sin x) \\ &= \sin x - \frac{1}{3} \sin^3 x + C \end{aligned}$$

*** Strategy for Evaluating $\int \sin^m x \cos^n x dx$**

(a) $n = 2k + 1$

$$\begin{aligned} \int \sin^m x \cos^{2k+1} x dx &= \int \sin^m x (\cos^2 x)^k \cos x dx \\ &= \int \sin^m x (1 - \sin^2 x)^k d(\sin x) \end{aligned}$$

(b) $m = 2k + 1$

$$\begin{aligned} \int \sin^{2k+1} x \cos^n x dx &= \int (\sin^2 x)^k \cos^n x \sin x dx \\ &= - \int (1 - \cos^2 x)^k \cos^n x d(\cos x) \end{aligned}$$

(c) $m = 2k$ and $n = 2k'$

$$\begin{aligned} \int \sin^{2k} x \cos^{2k'} x dx &= \int (\sin^2 x)^k (\cos^2 x)^{k'} dx \\ &= \int \left(\frac{1 - \cos 2x}{2} \right)^k \left(\frac{1 + \cos 2x}{2} \right)^{k'} dx \end{aligned}$$

(d) using $\sin x \cos x = \frac{1}{2} \sin 2x$

Exam 2 Find $\int \sin^5 x \cos^2 x dx$.

$$\begin{aligned} \text{(sol)} \quad \int \sin^5 x \cos^2 x dx &= \int (\sin^2 x)^2 \cos^2 x \sin x dx \\ &= - \int (1 - \cos^2 x)^2 \cos^2 x d(\cos x) \\ &= - \int (\cos^2 x - 2\cos^4 x + \cos^6 x) d(\cos x) \\ &= -\frac{1}{3} \cos^3 x + \frac{2}{5} \cos^5 x - \frac{1}{7} \cos^7 x + C \end{aligned}$$

Exam 3 Evaluate $\int_0^\pi \sin^2 x dx$.

$$\begin{aligned} \text{(sol)} \quad \int_0^\pi \sin^2 x dx &= \int_0^\pi \frac{1 - \cos 2x}{2} dx \\ &= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^\pi \\ &= \frac{\pi}{2} \end{aligned}$$

Exam 4 Find $\int \sin^4 x \, dx$.

$$\begin{aligned} \text{(sol)} \quad \int \sin^4 x \, dx &= \int (\sin^2 x)^2 \, dx \\ &= \int \left(\frac{1 - \cos 2x}{2} \right)^2 \, dx \\ &= \frac{1}{4} \int (1 - 2\cos 2x + \cos^2 2x) \, dx \\ &= \frac{1}{4} \int \left(1 - 2\cos 2x + \frac{1 + \cos 4x}{2} \right) \, dx \\ &= \frac{1}{4} \left(\frac{3}{2}x - \sin 2x + \frac{1}{8} \sin 4x \right) + C \end{aligned}$$

*** Strategy for Evaluating $\int \tan^m x \sec^n x \, dx$**

(a) $n = 2k \quad (k \geq 2)$

$$\begin{aligned} \int \tan^m x \sec^{2k} x \, dx &= \int \tan^m x (\sec^2 x)^{k-1} \sec^2 x \, dx \\ &= \int \tan^m x (1 + \tan^2 x)^{k-1} d(\tan x) \end{aligned}$$

(b) $m = 2k + 1$

$$\begin{aligned} \int \tan^{2k+1} x \sec^n x \, dx &= \int (\tan^2 x)^k \sec^{n-1} x \tan x \sec x \, dx \\ &= \int (\sec^2 x - 1)^k \sec^{n-1} x \, d(\sec x) \end{aligned}$$

(c) $m = 2k \quad \text{and} \quad n = 2k' + 1$

$$\int \tan^{2k} x \sec^{2k'+1} x \, dx = \int (\sec^2 x - 1)^k \sec^{2k'+1} x \, dx$$

Exam 5 Evaluate $\int \tan^6 x \sec^4 x \, dx$.

$$\begin{aligned} \text{(sol)} \quad \int \tan^6 x \sec^4 x \, dx &= \int \tan^6 x \sec^2 x \sec^2 x \, dx \\ &= \int \tan^6 x (1 + \tan^2 x) d(\tan x) \\ &= \int (\tan^6 x + \tan^8 x) d(\tan x) \\ &= \frac{1}{7} \tan^7 x + \frac{1}{9} \tan^9 x + C \end{aligned}$$

Exam 6 Find $\int \tan^5 \theta \sec^7 \theta d\theta$.

$$\begin{aligned} \text{(sol)} \quad \int \tan^5 \theta \sec^7 \theta d\theta &= \int (\tan^2 \theta)^2 \sec^6 \theta \tan \theta \sec \theta d\theta \\ &= \int (\sec^2 \theta - 1)^2 \sec^6 \theta d(\sec \theta) \\ &= \int (\sec^{10} \theta - 2\sec^8 \theta + \sec^6 \theta) d(\sec \theta) \\ &= \frac{1}{11} \sec^{11} \theta - \frac{2}{9} \sec^9 \theta + \frac{1}{7} \sec^7 \theta + C \end{aligned}$$

Exam 7 Find $\int \tan^3 x dx$.

$$\begin{aligned} \text{(sol)} \quad \int \tan^3 x dx &= \int \tan x \tan^2 x dx \\ &= \int \tan x (\sec^2 x - 1) dx \\ &= \int \tan x \sec^2 x dx - \int \tan x dx \\ &= \int \tan x d(\tan x) - \int \tan x dx \\ &= \frac{1}{2} \tan^2 x - \ln |\sec x| + C \end{aligned}$$

Exam 8 Find $\int \sec^3 x dx$.

$$\begin{aligned} \text{(sol)} \quad \int \sec^3 x dx &= \int \sec^2 x \sec x dx \\ &= \tan x \sec x - \int \tan^2 x \sec x dx \\ &= \tan x \sec x - \int (\sec^2 x - 1) \sec x dx \\ &= \tan x \sec x - \int \sec^3 x dx + \int \sec x dx \end{aligned}$$

$$\begin{aligned} \Rightarrow 2 \int \sec^3 x dx &= \tan x \sec x + \int \sec x dx \\ &= \tan x \sec x + \ln |\sec x + \tan x| + 2C \end{aligned}$$

$$\therefore \int \sec^3 x dx = \frac{1}{2} (\tan x \sec x + \ln |\sec x + \tan x|) + C$$

*** Trigonometric Substitution**

1. $\sqrt{a^2 - x^2} : x = a \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

$$\begin{aligned} \Rightarrow \sqrt{a^2 - x^2} &= \sqrt{a^2 - a^2 \sin^2 \theta} & \text{and} \quad dx &= a \cos \theta d\theta \\ &= \sqrt{a^2(1 - \sin^2 \theta)} \\ &= \sqrt{a^2 \cos^2 \theta} \\ &= a \cos \theta \end{aligned}$$

2. $\sqrt{a^2 + x^2} : x = a \tan \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$

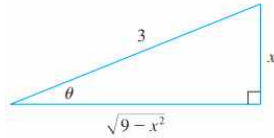
$$\begin{aligned} \Rightarrow \sqrt{a^2 + x^2} &= \sqrt{a^2 + a^2 \tan^2 \theta} & \text{and} \quad dx &= a \sec^2 \theta d\theta \\ &= \sqrt{a^2(1 + \tan^2 \theta)} \\ &= \sqrt{a^2 \sec^2 \theta} \\ &= a \sec \theta \end{aligned}$$

3. $\sqrt{x^2 - a^2} : x = a \sec \theta, \quad 0 \leq \theta < \frac{\pi}{2}, \pi \leq \theta < \frac{3\pi}{2}$

$$\begin{aligned} \Rightarrow \sqrt{x^2 - a^2} &= \sqrt{a^2 \sec^2 \theta - a^2} & \text{and} \quad dx &= a \sec \theta \tan \theta d\theta \\ &= \sqrt{a^2(\sec^2 \theta - 1)} \\ &= \sqrt{a^2 \tan^2 \theta} \\ &= a \tan \theta \end{aligned}$$

Exam 9 Evaluate $\int \frac{\sqrt{9-x^2}}{x^2} dx$.

(sol) Let $x = 3\sin\theta$, $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$



$$\cot\theta = \frac{\sqrt{9-x^2}}{x} \quad \text{and} \quad \theta = \sin^{-1}\left(\frac{x}{3}\right)$$

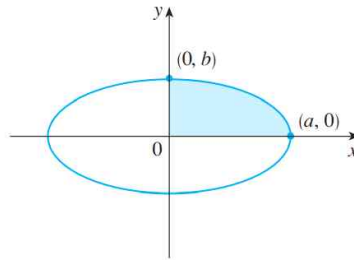
$$\Rightarrow dx = 3\cos\theta d\theta$$

$$\begin{aligned} \therefore \int \frac{\sqrt{9-x^2}}{x^2} dx &= \int \frac{\sqrt{9-9\sin^2\theta}}{9\sin^2\theta} 3\cos\theta d\theta \\ &= \int \frac{3\cos\theta}{9\sin^2\theta} 3\cos\theta d\theta \\ &= \int \cot^2\theta d\theta \\ &= \int (\csc^2\theta - 1) d\theta \\ &= -\cot\theta - \theta + C \\ &= -\frac{\sqrt{9-x^2}}{x} - \sin^{-1}\left(\frac{x}{3}\right) + C \end{aligned}$$

Exam 10 Find the area enclosed by the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$(\text{sol}) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \Rightarrow \quad y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$$



$\therefore$ The total area A is

$$\begin{aligned} A &= 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} \, dx \\ &= \frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} \, dx \end{aligned}$$

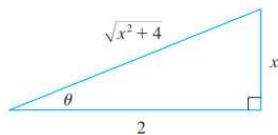
$$\text{Let } x = a \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$\Rightarrow dx = a \cos \theta \, d\theta, \quad x = 0 \rightarrow \theta = 0 \quad \text{and} \quad x = a \rightarrow \theta = \frac{\pi}{2}$$

$$\begin{aligned} \therefore A &= \frac{4b}{a} \int_0^{\pi/2} \sqrt{a^2 - a^2 \sin^2 \theta} \, a \cos \theta \, d\theta \\ &= 4ab \int_0^{\pi/2} \cos^2 \theta \, d\theta \\ &= 4ab \int_0^{\pi/2} \frac{1 + \cos 2\theta}{2} \, d\theta \\ &= 2ab \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\pi/2} \\ &= 2ab \left(\frac{\pi}{2} \right) \\ &= ab\pi \end{aligned}$$

Exam 11 Find $\int \frac{1}{x^2 \sqrt{x^2 + 4}} dx$.

(sol) $x = 2 \tan \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$



$$\csc \theta = \frac{\sqrt{x^2 + 4}}{x}$$

$$\Rightarrow dx = 2 \sec^2 \theta d\theta$$

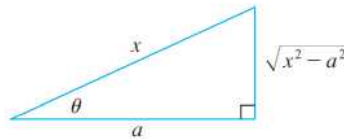
$$\begin{aligned} \therefore \int \frac{1}{x^2 \sqrt{x^2 + 4}} dx &= \int \frac{2 \sec^2 \theta d\theta}{4 \tan^2 \theta \sqrt{4 \tan^2 \theta + 4}} \\ &= \int \frac{2 \sec^2 \theta}{4 \tan^2 \theta 2 \sec \theta} d\theta \\ &= \frac{1}{4} \int \frac{\sec \theta}{\tan^2 \theta} d\theta \\ &= \frac{1}{4} \int \frac{1/\cos \theta}{\sin^2 \theta / \cos^2 \theta} d\theta \\ &= \frac{1}{4} \int \frac{\cos \theta}{\sin^2 \theta} d\theta \\ &= \frac{1}{4} \int \frac{1}{\sin^2 \theta} d(\sin \theta) \\ &= -\frac{1}{4} \frac{1}{\sin \theta} + C \\ &= -\frac{1}{4} \csc \theta + C \\ &= -\frac{\sqrt{x^2 + 4}}{4x} + C \end{aligned}$$

Exam 12 Find $\int \frac{x}{\sqrt{x^2+4}} dx$.

(sol)
$$\begin{aligned}\int \frac{x}{\sqrt{x^2+4}} dx &= \frac{1}{2} \int (x^2+4)^{-\frac{1}{2}} d(x^2+4) \\ &= \frac{1}{2} 2(x^2+4)^{\frac{1}{2}} + C \\ &= \sqrt{x^2+4} + C\end{aligned}$$

Exam 13 Evaluate $\int \frac{dx}{\sqrt{x^2-a^2}}$, where $a > 0$.

(sol) Let $x = a \sec \theta$, $0 \leq \theta < \frac{\pi}{2}$, $\pi \leq \theta < \frac{3\pi}{2}$



$$\tan \theta = \frac{\sqrt{x^2-a^2}}{a} \quad \text{and} \quad \sec \theta = \frac{x}{a}$$

$$\Rightarrow dx = a \sec \theta \tan \theta d\theta$$

$$\begin{aligned}\therefore \int \frac{dx}{\sqrt{x^2-a^2}} &= \int \frac{a \sec \theta \tan \theta}{\sqrt{a^2 \sec^2 \theta - a^2}} d\theta \\ &= \int \frac{a \sec \theta \tan \theta}{a \tan \theta} d\theta \\ &= \int \sec \theta d\theta \\ &= \ln |\sec \theta + \tan \theta| + C' \\ &= \ln \left| \frac{x}{a} + \frac{\sqrt{x^2-a^2}}{a} \right| + C' \\ &= \ln |x + \sqrt{x^2-a^2}| - \ln a + C' \\ &= \ln |x + \sqrt{x^2-a^2}| + C\end{aligned}$$

where $C = C' - \ln a$.

Exam 14 Find $\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(4x^2+9)^{3/2}} dx$.

(sol) Let $x = \frac{3}{2} \tan \theta$, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

$\Rightarrow dx = \frac{3}{2} \sec^2 \theta d\theta$, $x=0 \rightarrow \theta=0$ and $x = \frac{3\sqrt{3}}{2} \rightarrow \theta = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$

$$\begin{aligned} \therefore \int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(4x^2+9)^{3/2}} dx &= \int_0^{\frac{\pi}{3}} \frac{\frac{27}{8} \tan^3 \theta}{(9 \tan^2 \theta + 9)^{3/2}} \frac{3}{2} \sec^2 \theta d\theta \\ &= \int_0^{\frac{\pi}{3}} \frac{\frac{27}{8} \tan^3 \theta}{27 \sec^3 \theta} \frac{3}{2} \sec^2 \theta d\theta \\ &= \frac{3}{16} \int_0^{\frac{\pi}{3}} \frac{\tan^3 \theta}{\sec \theta} d\theta \\ &= \frac{3}{16} \int_0^{\frac{\pi}{3}} \frac{\sin^3 \theta}{\cos^2 \theta} d\theta \\ &= \frac{3}{16} \int_0^{\frac{\pi}{3}} \frac{1 - \cos^2 \theta}{\cos^2 \theta} \sin \theta d\theta \\ &= \frac{3}{16} \left[\int_0^{\frac{\pi}{3}} \frac{\sin \theta}{\cos^2 \theta} d\theta - \int_0^{\frac{\pi}{3}} \sin \theta d\theta \right] \\ &= \frac{3}{16} \left[- \int_0^{\frac{\pi}{3}} \frac{1}{\cos^2 \theta} d(\cos \theta) - \int_0^{\frac{\pi}{3}} \sin \theta d\theta \right] \\ &= \frac{3}{16} \left[\frac{1}{\cos \theta} + \cos \theta \right]_0^{\frac{\pi}{3}} \\ &= \frac{3}{16} \left(2 + \frac{1}{2} - 1 - 1 \right) \\ &= \frac{3}{32} \end{aligned}$$

(Exam) Evaluate $\int \frac{x}{\sqrt{3-2x-x^2}} dx$.

$$\text{(sol)} \quad \int \frac{x}{\sqrt{3-2x-x^2}} dx = \int \frac{x}{\sqrt{4-(x+1)^2}} dx$$

$$\text{Let } x+1=2\sin\theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$\Rightarrow \cos\theta = \sqrt{1-\sin^2\theta} = \frac{\sqrt{3-2x-x^2}}{2}, \quad \theta = \sin^{-1}\left(\frac{x+1}{2}\right)$$

$$\text{and } dx = 2\cos\theta d\theta$$

$$\begin{aligned} \therefore \int \frac{x}{\sqrt{3-2x-x^2}} dx &= \int \frac{2\sin\theta-1}{\sqrt{4-4\sin^2\theta}} 2\cos\theta d\theta \\ &= \int \frac{2\sin\theta-1}{2\cos\theta} 2\cos\theta d\theta \\ &= \int (2\sin\theta-1) d\theta \\ &= -2\cos\theta - \theta + C \\ &= -\sqrt{3-2x-x^2} - \sin^{-1}\left(\frac{x+1}{2}\right) + C \end{aligned}$$